# Project Scope

# GenAI Co-Build Scope Agreement

# Cloud AI FDE Team

## Project Name

Support Ticket Router & Assistant

## Engagement Type

Onboarding Project 4 (Customer Support Domain)

## Staging Environment

100% Self-Contained in Argolis Sandbox

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## Metadata

- **Customer Name**: Global Retail Hub (Simulated)
- **Agreement Date**: June 13, 2026
- **Agreement Status**: Draft
- **Customer Sponsor**: [Global Retail VP of Customer Experience]
- **Google Sponsor**: GenAI FDE Manager (M1)
- **Customer Lead**: [Global Retail Support Platform Lead]
- **Google Lead**: FDE Pod Lead / Onboarding Buddy

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## Company Overview

Global Retail Hub is a fast-growing international e-commerce platform. The company handles millions of customer inquiries daily across chat, email, and social media. To manage this volume, the support engineering team requires an automated ticket routing and assistance

system that can analyze customer sentiment, resolve common FAQs using grounded knowledge, and escalate complex or frustrated tickets to human agents instantly.

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# Project Overview

## Support Ticket Router & Assistant (AI Sandbox Initiative)

Lay the underlying foundation for customer support agentic capabilities by developing a secure, self-contained AI Support Operations Sandbox.

This engagement focuses on establishing a secure, scalable "AI Sandbox" inside the onboarding FDE's allocated Argolis Project. It empowers support engineers to build a sentiment-aware router and automated FAQ assistant. The project will bridge the gap between customers demanding instant, personalized replies and the operational need to route tickets accurately while respecting API rate limits and storing conversation states securely.

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# Project Scope

| **Scope / Deliverable** | **Responsible Party** |
|---|---|
| Design of the Routing Architecture: FastAPI-based backend that classifies message sentiment and routes requests to automated RAG lookup or human queues. | Google FDE (Noogler) |
| Ingestion Pipeline & Grounding Index: Ingestion of Kaggle Support Customer Support Twitter dataset into Agent Platform (fka Vertex AI) Agent Search. | Google FDE (Noogler) |
| Database State Management: Configuration of BigQuery / Cloud SQL to persist ticket status, sentiment history, and chat logs. | Google FDE (Noogler) |
| Reliability Implementation: Integration of a Circuit-Breaker pattern with exponential backoff on external API connections (FastAPI middleware). | Google FDE (Noogler) |

| Scope / Deliverable | Responsible Party |
| --- | --- |
| Safety & Safety Integrations: Redaction of user PII (emails, order numbers) before model processing using Model Armor. | Google FDE (Noogler) |
| Automated Verification: pytest suite with mocked RAG Engine and Cloud SQL interfaces, achieving >80% code coverage. | Google FDE (Noogler) |

### Out of Scope

- Integration with actual production social media APIs (Twitter/Facebook live connections).
- Live payment processing or order cancellation capabilities (read-only mock database checks are in scope).
- Deployments beyond the allocated, self-contained Argolis Sandbox environment.

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## User Journey

1. **Inbound Message**: A customer submits a message (e.g., "Where is my order #12345? It was supposed to arrive yesterday and I'm very frustrated!").
2. **Sentiment Analysis**: The FastAPI backend receives the request and invokes Gemini 3.5 Flash to analyze sentiment and extract entities (Order ID: 12345, Sentiment: Angry/Urgent).
3. **State Persistence**: The ticket is saved in Cloud SQL with status "Active", sentiment "Angry", and assigned to "Escalation Queue".
4. **Routing Logic**:
    - *If Angry/Urgent*: The router bypasses automatic RAG reply and assigns the ticket to a human queue, returning a calming acknowledgement.
    - *If Neutral/Factual*: The router queries Agent Platform (fka Vertex AI) Agent Search for matching FAQs or similar resolved cases and returns a grounded answer.
5. **Circuit Breaker Fallback**: If the Agent Platform (fka Vertex AI) Agent Search is overloaded (throwing HTTP 429), the circuit breaker opens, and the system gracefully routes the ticket to the human queue without crashing.
6. **Ground Truth Citation**: Grounded replies contain citations pointing to the mock FAQ source, displayed in the split-pane frontend interface.

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# Objectives

## Business Objectives

- **Improve CSAT (Customer Satisfaction)**: Prioritize angry or frustrated customers by routing them to human agents instantly, reducing response time for critical issues.
- **Deflect Common Queries**: Automate replies for standard factual questions (shipping policies, returns), deflecting >40% of standard ticket volume.
- **System High-Availability**: Ensure support systems remain functional even during high-volume events or GenAI quota outages.

## Technical Objectives

- **Stateful Orchestration**: Implement BigQuery / Cloud SQL to manage ticket states and conversation history.
- **Resilient RAG Retrieval**: Integrate Agent Platform (fka Vertex AI) Agent Search for scalable retrieval over customer support datasets.
- **Circuit-Breaker Pattern**: Enforce FastAPI middleware that detects API rate limits (HTTP 429) and falls back to deterministic queues, protecting backend stability.

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# Timeline

| Sprint | Phase | User Stories | Tasks & Owners |
|---|---|---|---|
| **Sprint 1** | **Tactical (Discovery)** | As a developer, I want to initialize my git repo, provision Cloud SQL, and configure Agent Platform (fka Vertex AI) Agent Search. | Initialize Repo, configure Cloud SQL, configure RAG Engine. (FDE Noogler) |
| **Sprint 2** | **Tactical (Grounding)** | As an engineer, I want to clean the Twitter support dataset and index it into Agent Platform (fka Vertex AI) Agent Search. | Write dataset cleaning scripts, index files. (FDE Noogler) |

| Sprint | Phase | User Stories | Tasks & Owners |
|---|---|---|---|
| **Sprint 3** | **Strategic (Orchestration)** | As a developer, I want to build the FastAPI router, implement sentiment classification, and connect Cloud SQL persistence. | Develop routing logic, integrate sentiment API, build DB models. (FDE Noogler) |
| **Sprint 4** | **Strategic (Reliability & Safety)** | As a developer, I want to implement the circuit-breaker pattern, integrate Model Armor, and configure OpenTelemetry. | Implement FastAPI circuit-breaker middleware, configure Model Armor and OTEL. (FDE Noogler) |
| **Sprint 5** | **Strategic (Deployment)** | As an engineer, I want to containerize my app with Docker, script my Terraform HCL, and deploy to Google Cloud Run. | Write Dockerfile, Terraform, and execute deployment. (FDE Noogler) |
| **Sprint 6** | **Disengagement** | As a builder, I want to run my Product Readiness Review (PRR) readout, sanitize my code, and transition assets. | Run PRR readout with Lead, upload clean code to FDE shared repository. (FDE Noogler) |

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# Technical Requirements

## Onboarding (Model A - In-Environment)

- Provisioning occurs inside the FDE's allocated Argolis Project sandbox.
- Access to development organizations and tools (GitHub, etc.).
- Development completed using JetSki / AGY 2.0 with Gemini 3.5 Flash assistance.

### Data Sources

- **Primary Corpus**: Customer Support dataset (curated Kaggle CSV).
- **Storage**: Ingested into Agent Platform (fka Vertex AI) Agent Search. Ticket logs and history stored in BigQuery / Cloud SQL.
- **Data Policy**: Simulated public tweets. PII (emails, credit cards) masked via Model Armor.

### Integration Points

- **BigQuery / Cloud SQL**: For managing database sessions and state.
- **Agent Platform (fka Vertex AI) Agent Search API**: For FAQ document retrieval.

### Technology Requirements

- **Runtime**: Python 3.11+ / FastAPI
- **Frontend**: React 18+ / TypeScript / Tailwind CSS
- **Orchestration**: Google ADK / AGY SDK
- **Models**: Gemini 2.5 Flash (for sentiment routing), Gemini 2.5 Pro (for complex reasoning/FAQ compilation)
- **Infrastructure**: Google Cloud Run, Cloud SQL (PostgreSQL), Cloud Storage, BigQuery, Secret Manager, VPC Service Controls, OpenTelemetry, Cloud Trace

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## Working Methodology and Environment Requirements

This project follows **Model A: In-Environment Development**.

- **Development Environment**: Complete development staged inside the secure Argolis sandbox.
- **Tooling**: Access to JetSki IDE and Gemini 3.5 Flash.
- **Infrastructure**: Dedicated GCP Dev Project (`fde-support-sandbox-dev`) with Vertex AI, Cloud Run, Cloud SQL, GCS, BigQuery, and Secret Manager active.
- **Identity/IAM**: Enforced Least-Privilege IAM roles for the service accounts.
- **Data Policy**: Zero production client PII.

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## Team, Cadence, and Logistics

- **Manager (M1)**: General oversight, milestone gates, and PRR escalation.

- **Pod Lead / Buddy**: Strategic advisor, code reviews, and PR approvals (50-70% technical contribution).
- **Core Builder (Noogler)**: Tactical owner of the end-to-end SDLC and final deliverable.
- **Daily Rhythm**: 15-minute morning standup. Weekly status review and code walkthroughs.
- **Communication Channel**: Shared Google Chat Space (`fde-onboard-ticket-router`).

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# Success Metrics & Transition Plan

## Success Criteria (The "Definition of Done")

- **Functional Verification**: Average score of 2-Competent across all FDE grading categories.
    - 100% of "Angry" or "Urgent" classified tickets successfully bypassed to human queues.
    - Circuit breaker opens after 3 consecutive HTTP 429 errors, redirecting traffic safely to human queues within <500ms overhead.
    - >=80% code coverage on FastAPI backend router logic.
- **Operational Verification**: Containerized via Docker, provisioned entirely via Terraform, and successfully deployed to Google Cloud Run with Cloud SQL database integration.
- **Observability**: Completed OpenTelemetry and Cloud Trace setup, exporting logs, DB latencies, and circuit-breaker states.

## Transition Plan

- **Platform Extraction**: Upon successful completion of Milestone 5, the Noogler enters the mandated 1-week Squad Cooldown.
- **Sanitation & Harvesting**: The Noogler generalizes the routing configurations, removes database credentials, and pushes the reusable components to the FDE central repository (`//FDE_shared/agent_engineering/`).
- **Asset Registration**: The containerized mock environment is published to the Demo Library and transitioned to the demo operations team.